

FININ: paper and prototype

A step-by-step map of the authors' model and the working student-scale implementation.



TIME ALIGNMENT

Paper: market + news through day d (or a t-day lookback) -> whether $close(d+1) > close(d)$.
Prototype: headlines dated n -> SPY close on next session d -> whether close on following session q > close(d).

Paper: [1] Figure 2 and Sections 3-5. Prototype: [2] source modules listed in project architecture guide.

What changed - and why it matters

The same next-session direction question is tested with different information and a smaller protocol.

ASPECT	PAPER [1]	PROTOTYPE [2]
Market and years	S&P 500 and NASDAQ 100 indices; daily data from 2003-2018.	SPY ETF proxy; NIFTY-covered news dates from 2010-2020.
News scale	Reuters TRNA: over 2.7 million items; 24-1,953 items per day in Appendix A.	NIFTY: 2,111 dated rows; 2,107 usable examples; 33,673 selected headline slots.
News sentiment	TRNA probabilities for positive, neutral or negative influence on named instruments.	TinyBERT-generated positive/neutral/negative probabilities for headline text.
Market input	Six price/volume fields; market descriptions and related company names.	Six normalized SPY channels; one fixed market description, no constituent list.
News timing	Market and news through day d; lookbacks of 1, 3, 5, 10 and 20 days.	No item timestamps: previous day's news + market through forecast close; one input day.
News selection	Designed to use all available daily TRNA news items.	At most 16 exact-deduplicated headlines per day, selected by a fixed hash.
Validation	Ten sliding 500-day windows; chronological 8:1:1 split in each.	One chronological split: 1,475 train / 315 valid / 317 test; 4 boundary exclusions.
Reported metrics	Accuracy, profit-and-loss and Sharpe ratio; strongest claim is Sharpe improvement.	Accuracy, balanced accuracy and log loss; no trading or Sharpe calculation.

INTERPRETATION

Architecture reproduced; Sharpe claim untested. On 317 test dates: 58.4% accuracy vs 58.0% always-up, with 50.4% balanced accuracy.

[1] Wang, Cohen and Ma, Findings of EMNLP 2024, Sections 3-6, Figure 2, Tables 2-3, Appendix A.

[2] This project's data_preparation.md, results.md and documented implementation (snapshot: 30 Sep 2026).